Mason Fleming

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EDUCATION

Purdue University, Bachelor of Science in Computer Engineering — Expected Graduation Date: May 2026
Relevant Curriculum: Computer Design and Prototyping, ASIC Design Lab, Microprocessor Systems & Interfacing, Advanced C Programming, Compilers & Translation Engineering, Radiation Effects on Transistors, Changing the World With Chips.

EMPLOYMENT EXPERIENCE

Purdue University School of Electrical Engineering (Dr. Peter Bermel), Undergraduate SCALE Researcher — 1/24 – Present

- Designed and fabricated PCB test platforms for deployment in a subcritical nuclear pile, integrating radiation-sensitive ICs to monitor bit flips, drain current transients, and system resets under neutron and gamma exposure up to 10^6–10^7 particles/cm², enabling fault characterization in hostile environments.
- Developed Python and C-based TCAD simulations (Sentaurus) to model GAAFET and FinFET behavior under heavy-ion, neutron, and gamma radiation, demonstrating up to ~12% lower transient drain current perturbation in GAAFETs compared to FinFETs during single-event strikes.
- Quantified the impact of transistor geometry on radiation susceptibility, showing that channel widths <4 nm significantly increased interface trap density, while optimized nanosheet geometries reduced radiation-induced current variation by ~10^{-10} % per ion strike.
- Constructed physics based radiation environments using Geant4, simulating proton, neutron, and gamma spectra from space and reactor sources; validated models against published ISS and fission-spectrum data to ensure cross-environment reliability of device behavior predictions.

University of Texas at Austin Cockrell School of Engineering (Dr. David DiCarlo), Undergraduate Research Intern — 6/23 – 8/23

- Designed and programmed control firmware for a programmable sand-distribution apparatus, enabling repeatable construction of heterogeneous subsurface bedforms used to experimentally evaluate CO_2 and hydrocarbon trapping efficiency across multiple geologic patterns.
- Analyzed multiphase flow experiments in MATLAB, quantifying the effect of bead size, saturation, and bedform geometry on gas retention; results showed only ~66% non-wetting phase trapping for the tested pattern, with breakthrough occurring at ~30 mL injected volume, indicating poor long-term storage viability.
- Processed pixel-wise saturation data from optical transmission measurements, demonstrating that saturation was not preserved during redistribution, and identifying pattern contrast as a limiting factor for trapping performance compared to higher-efficiency reference geometries.

University of Texas at Austin Governmental Affairs, Calculus Grader — 9/22 – 4/23

- Collaborated with a team to grade calculus exams accurately and efficiently, ensuring consistent and fair evaluations.
- Completed professional development workshops on teamwork, constructive critique, and effective communication.

PROJECT EXPERIENCE

Purdue IEEE EMBS, Club Chair, West Lafayette, IN — 1/25 – Present

- Expanded the organization from 3 to ~20 active members in one semester by increasing outreach, collaborating with other IEEE committees, and running consistent weekly project development meetings that kept members engaged by providing one-on-one support and technical guidance as needed.
- Designed onboarding programs covering Arduino firmware development, timing concepts, pin configuration, debugging, GitHub workflows, circuit design, PCB design, pinout interpretation, and soldering allowing members to independently design, program, and build functional embedded system prototypes.
- Developed biomedical monitoring systems, including a sleep apnea detection device and a neurodegenerative disease (e.g., Alzheimer's) monitoring device, integrating SpO_2, ECG, and microphone sensors for physiological signal acquisition and processing.
- Organized and led an EMG controlled prosthetic claw workshop, where students designed, built, and demonstrated a functional appendage by using electrical muscle contraction signals.

Purdue MIND, Hardware Designer, West Lafayette, IN — 4/24 – 1/25

- Created a circuit to amplify photodiode signals, filter out noise, and transmit fNIRS signals through Bluetooth for detection of deoxygenated blood in regions of brain injury.
- Delivered presentations on research findings and design parameters to hardware team and taught biomedical engineering students in basic hardware principles for aid in their designs.

SKILLS

C, C++, MATLAB, Python, RISC-V, SolidWorks, RTL Design (Verilog/SystemVerilog), TCAD, CAD, KiCad, Soldering